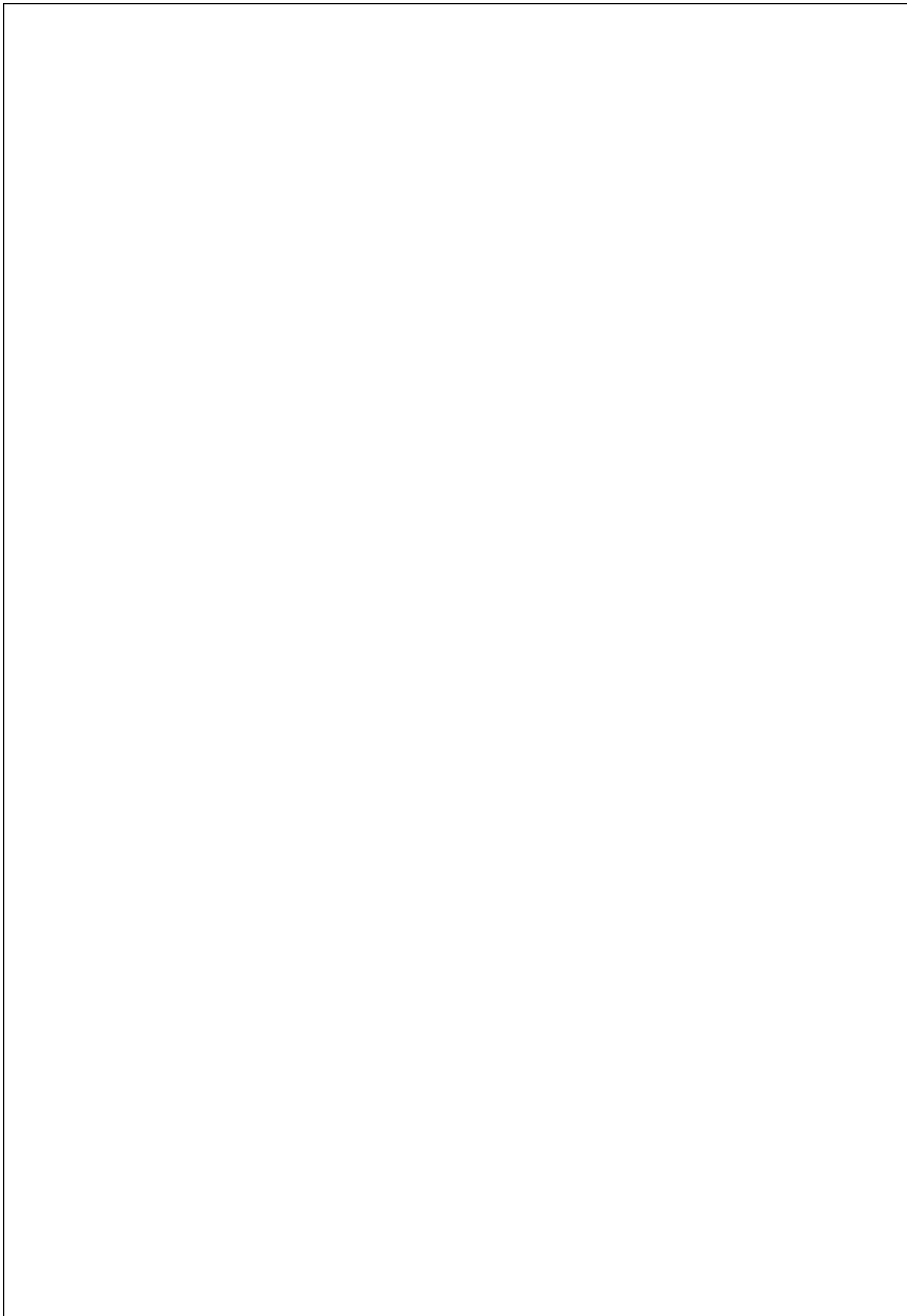




Ex.No.9 Use Process Explorer to identify suspicious processes

Objective

To conduct an in-depth behavioral and structural analysis of active operating system processes using Sysinternals Process Explorer; to evaluate hierarchical parent-child lineages, cryptographic digital signatures, and execution paths; to leverage cloud-based threat intelligence for automated multiengine malware identification; and to successfully isolate, contain, and terminate unauthorized, masquerading, or anomalous processes to secure system integrity.

Description

Process Explorer is a powerful Windows tool that provides detailed information about running processes, allowing users to monitor, troubleshoot, and detect suspicious activities. It can be used to detect potential malware or harmful processes by analyzing various aspects of running applications. Here's a step-by-step guide on how to use Process Explorer to identify suspicious processes on your system:

Step 1: Download and Set Up Process Explorer

1. Download Process Explorer:

- Go to the official Microsoft Sysinternals website and download Process Explorer: [Download Process Explorer](#).

2. Extract the Program:

- Once downloaded, extract the ZIP file to a folder.

3. Run Process Explorer:

- Open the folder where you extracted the files and run procexp64.exe (64-bit) or procexp.exe (32-bit) as Administrator by right-clicking and selecting "Run as Administrator."

Step 2: Familiarize Yourself with the Interface

When Process Explorer starts, you'll see an interface that lists all running processes, showing their hierarchy, memory usage, CPU usage, and much more.

1. **Process Tree:** ○ The main window shows a tree of all running processes, with each child process listed under its parent.
 - **Processes are color-coded to indicate their status:**
 - Pink: Suspended processes.
 - Light Blue: Processes running under the same user as you.
 - Dark Blue: Services or processes running under system accounts.

- Green: New processes (appearing and disappearing quickly may be suspicious).

- Red: Processes that have just exited.

2. Columns:

- **Various columns show information such as Process ID (PID), CPU usage, Memory usage, and more.**
-

Step 3: Identify Suspicious Processes

Now that you are familiar with the interface, follow these steps to investigate potentially suspicious processes:

1. Look for Unfamiliar Processes:

- Scan the list of running processes and identify any names that you don't recognize. Some malware disguises itself with legitimate-looking names, but odd names can be a red flag.
- Legitimate processes should typically be from trusted sources (e.g., Microsoft, Intel, Adobe, etc.).

2. Verify Digital Signatures:

- Right-click any suspicious process and select Properties.
- Go to the Image tab and check if the process has a valid Digital Signature.
- Legitimate applications from trusted developers will have valid signatures. Processes without signatures or with invalid signatures are suspicious.

3. Check the Path of the Process:

- In the Properties window of the suspicious process, look under the Image tab for the Path.
- Verify if the process is located in a legitimate directory. For example, legitimate system processes should reside in C:\Windows\System32.
- If the process is running from a temporary folder, user folder, or a random directory, it might be malware.

4. Monitor CPU, Memory, and Disk Usage:

- Look at the CPU, Memory, and Disk usage columns.
- A process that is using an unusual amount of resources without explanation could be malicious. For example, high CPU usage on a process like svchost.exe can indicate malware.

5. Check Process Description and Company Name:

- Look at the Description and Company Name columns in Process Explorer.
- Legitimate processes usually provide clear descriptions and are tied to well-known companies.
- Lack of information or suspicious company names should prompt further investigation.

6. Check the Process Network Activity:

- Some malware communicates with external servers. You can inspect a process's network connections:
 - Right-click on the process and select Properties, then go to the TCP/IP tab to see network activity associated with the process.
 - Unexpected or unexplained external connections can indicate a malicious process.
-

Step 4: Perform a Google Search on Suspicious Processes

If you find a process that looks suspicious or unfamiliar:

1. Search online: Perform a quick Google search using the process name (e.g., processname.exe) to find out whether the process is known to be safe or malicious.
 2. Check Malware Databases: Use online databases such as VirusTotal or ProcessLibrary to verify whether the process is known for being part of malware.
-

Step 5: Kill or Suspend Suspicious Processes If

you find a process that you believe is malicious:

1. **Kill the Process:** ○ Right-click the suspicious process and select Kill Process to terminate it immediately.
 2. **Suspend the Process:**
 - If you are unsure and want to prevent the process from causing damage while you investigate further, right-click and select Suspend.
 3. **Remove the Source File:**
 - Go to the location of the file (using the Path from the Properties window) and delete the executable file if confirmed to be malware.
 - Be cautious, as some malware might prevent you from deleting it. In such cases, you may need to boot into Safe Mode or use a specialized tool.
-

Step 6: Scan Your System

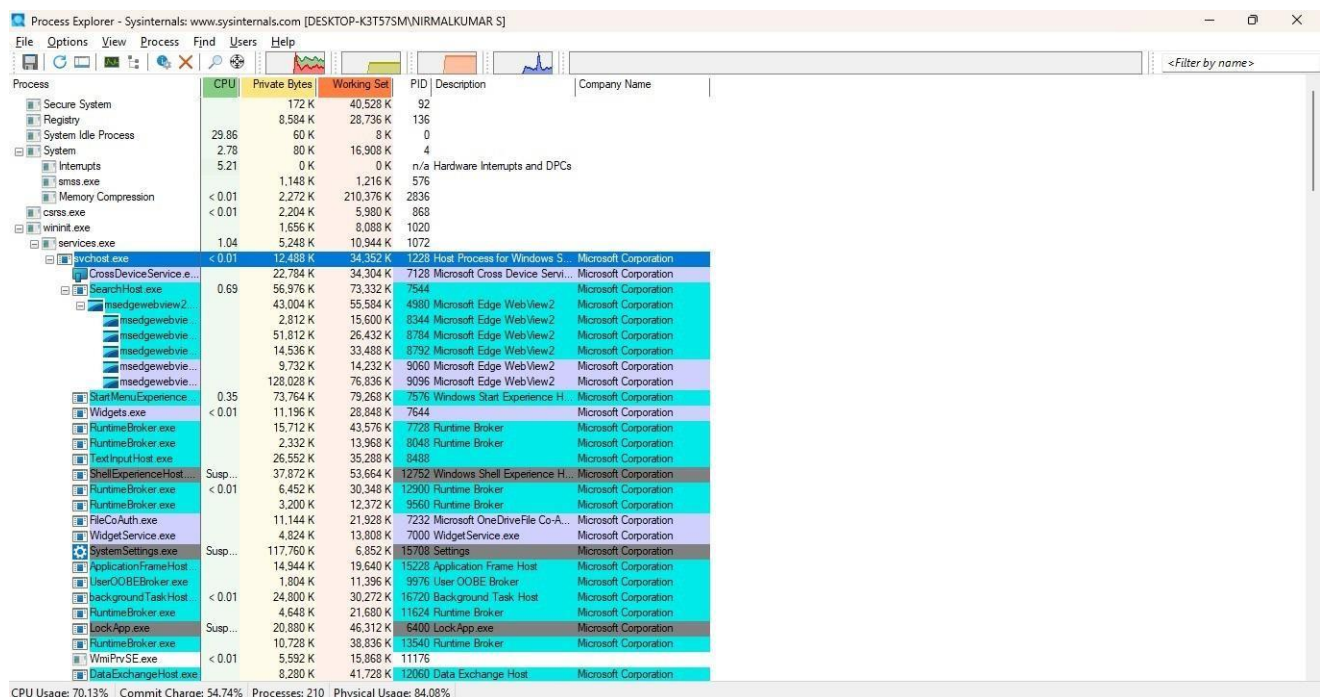
After terminating any suspicious processes, it's a good idea to:

- Run a full antivirus scan using trusted antivirus software.
- Use Malware Removal Tools like Malwarebytes or Windows Defender to perform an in-depth scan of your system.

Example of Identifying a Suspicious Process

1. You notice a process called randomname123.exe using an unusually high amount of CPU.
2. Right-click it, select Properties, and find it is running from a suspicious folder, not the system folder.
3. You check the TCP/IP tab and see it is making external connections to unknown IP addresses.
4. A quick search shows that this process is often associated with a specific type of malware.
5. You kill the process and remove the executable file from the system.

Output:



Process Explorer - Sysinternals: www.sysinternals.com [DESKTOP-K3T57SM\NIRMALKUMAR S]

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		172 K	40,528 K	92		
Registry		8,584 K	28,736 K	136		
System Idle Process	29.86	60 K	8 K	0		
System	2.78	80 K	16,908 K	4		
Interrupts	5.21	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,148 K	1,216 K	576		
Memory Compression	< 0.01	2,272 K	210,376 K	2836		
csrss.exe	< 0.01	2,204 K	5,980 K	868		
wininit.exe		1,656 K	8,088 K	1020		
services.exe	1.04	5,248 K	10,944 K	1072		
svchost.exe	< 0.01	12,488 K	34,352 K	1228	Host Process for Windows S...	Microsoft Corporation
CrossDeviceService.e...		22,784 K	34,304 K	7128	Microsoft Cross Device Servi...	Microsoft Corporation
SearchHost.exe	0.69	56,976 K	73,332 K	7544		Microsoft Corporation
msedgewebview2		43,004 K	55,584 K	4980	Microsoft Edge WebView2	Microsoft Corporation
msedgewebvie		2,812 K	15,600 K	8344	Microsoft Edge WebView2	Microsoft Corporation
msedgewebvie		51,812 K	26,432 K	8784	Microsoft Edge WebView2	Microsoft Corporation
msedgewebvie		14,536 K	33,488 K	8792	Microsoft Edge WebView2	Microsoft Corporation
msedgewebvie...		9,732 K	14,232 K	9060	Microsoft Edge WebView2	Microsoft Corporation
msedgewebvie...		128,028 K	76,836 K	9096	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperience...	0.35	73,764 K	79,268 K	7576	Windows Start Experience H...	Microsoft Corporation
Widgets.exe	< 0.01	11,196 K	28,848 K	7644		Microsoft Corporation
RuntimeBroker.exe		15,712 K	43,576 K	7728	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,332 K	13,968 K	8048	Runtime Broker	Microsoft Corporation
TextInputHost.exe		26,552 K	35,288 K	8488		Microsoft Corporation
ShellExperienceHost...	Susp...	37,872 K	53,664 K	12752	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe	< 0.01	6,452 K	30,348 K	12900	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		3,200 K	12,372 K	9560	Runtime Broker	Microsoft Corporation
FileCoAuth.exe		11,144 K	21,928 K	7232	Microsoft OneDriveFile Co-A...	Microsoft Corporation
WidgetService.exe		4,824 K	13,808 K	7000	WidgetService.exe	Microsoft Corporation
SystemSettings.exe	Susp...	117,760 K	6,852 K	15708	Settings	Microsoft Corporation
ApplicationFrameHost		14,944 K	19,640 K	15228	Application Frame Host	Microsoft Corporation
UserOOBEBroker.exe		1,804 K	11,396 K	9976	User OOBEBroker	Microsoft Corporation
backgroundTaskHost	< 0.01	24,800 K	30,272 K	16720	Background Task Host	Microsoft Corporation
RuntimeBroker.exe		4,648 K	21,680 K	11624	Runtime Broker	Microsoft Corporation
LockApp.exe	Susp...	20,880 K	46,312 K	6400	LockApp.exe	Microsoft Corporation
RuntimeBroker.exe		10,728 K	38,836 K	13540	Runtime Broker	Microsoft Corporation
WmiPrvSE.exe	< 0.01	5,592 K	15,868 K	11176		
DataExchangeHost.exe		8,280 K	41,728 K	12060	Data Exchange Host	Microsoft Corporation

CPU Usage: 70.13% | Commit Charge: 54.74% | Processes: 210 | Physical Usage: 84.08%

Figure 48

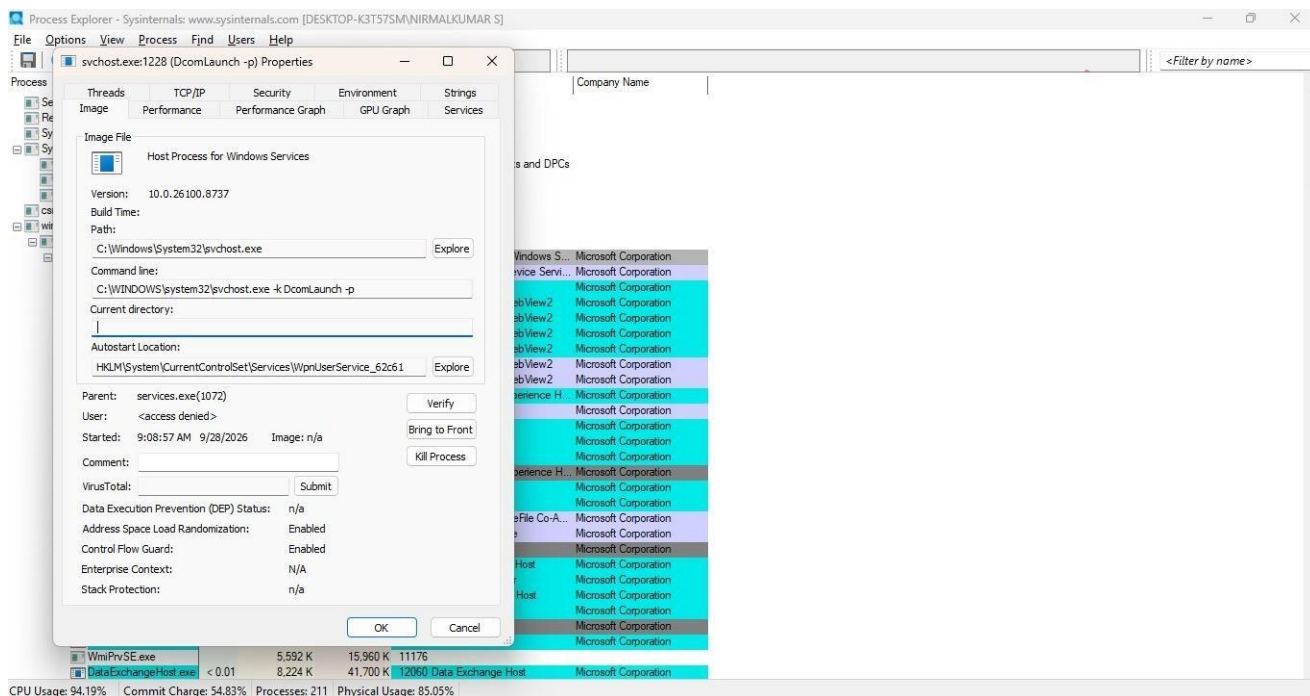


Figure 49

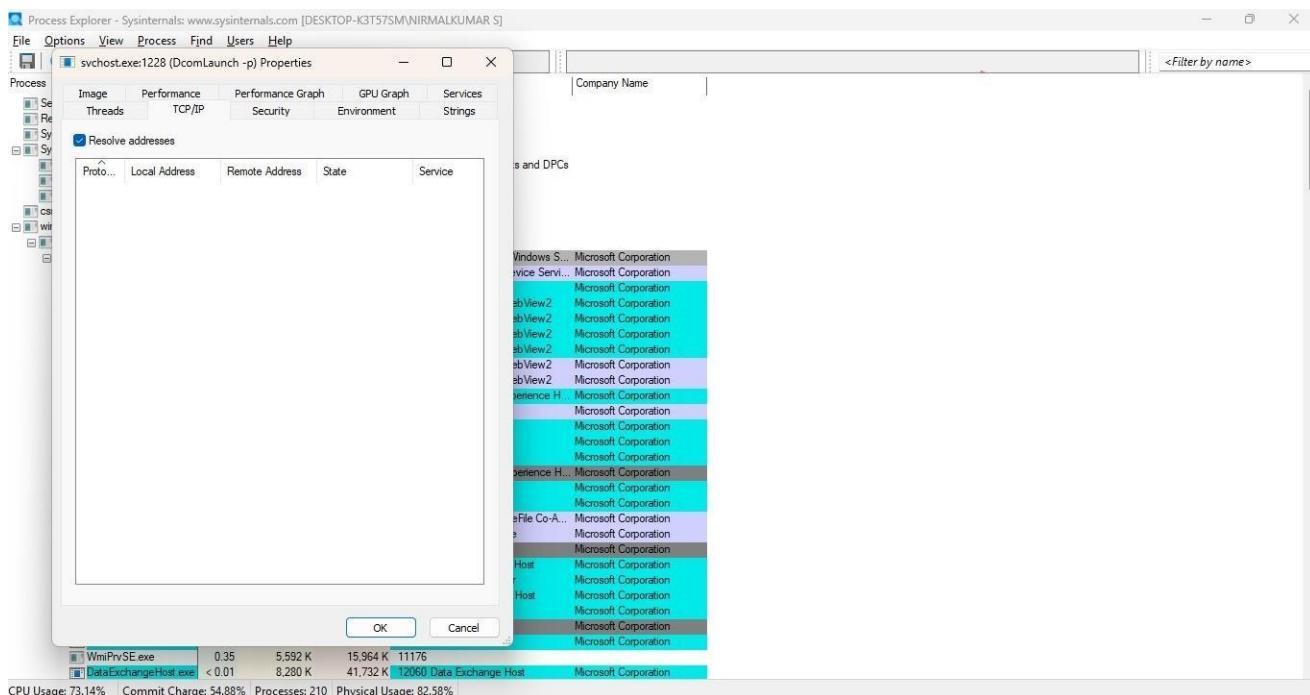


Figure 50

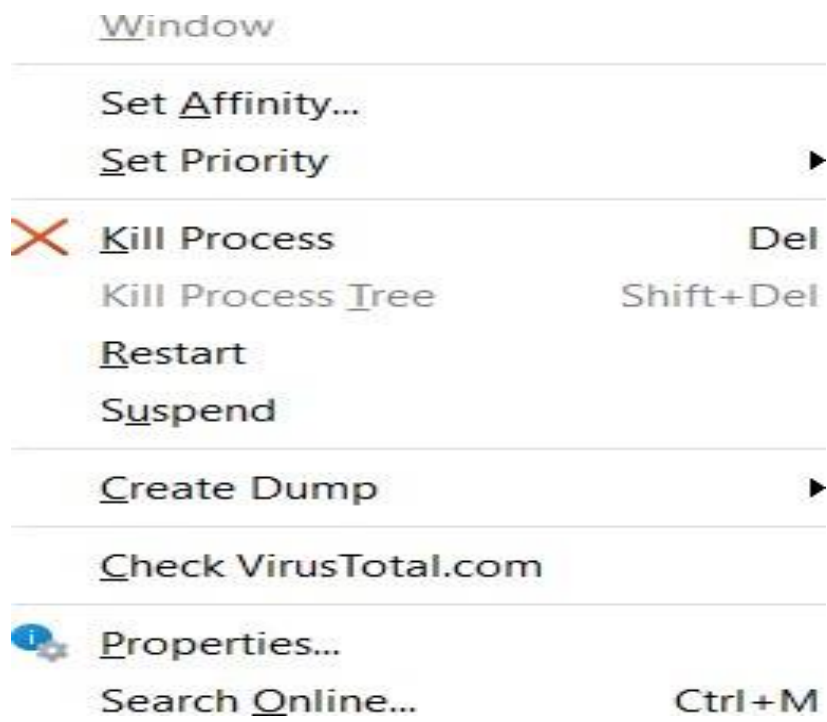


Figure 51

